LESSON 09 · CHEAT SHEET

pandas I: DataFrames

IN AND OUT

```
import pandas as pd
pd.set_option("display.width", 120)
df = pd.read_csv("data/file.csv")
df = pd.read_excel("data/book.xlsx", sheet_name="Payrol
df.to_csv("my_work/out.csv", index=False)
df.to_excel("my_work/out.xlsx", index=False)
```

LOOK AT IT

```
df.shape          (200, 12)  # rows, columns
df.columns        # column names
df.head()  df.head(3)  df.tail(3)
df.info()         # types + missing values
df.describe()     # count mean std min 25% ...
len(df)           200
```

COLUMNS

```
df["annual_salary"]            # one column (Series)
df[["first_name", "last_name"]] # several (list!)
df["annual_salary"].mean()     # .median() .max()
df["annual_salary"].sum()      # .min() .std()
df["department"].unique()      # distinct values
df["department"].nunique()     # how many distinct
df["new"] = df["a"] / df["b"]  # add a column
df["name"] = df["first"] + " " + df["last"]
df = df.drop(columns=["tmp"])
df = df.rename(columns={"old": "new"})
```

FILTER ROWS (AUTOFILTER)

```
df[df["annual_salary"] > 40000]
df[(df["dept"] == "Sales") & (df["hours"] < 37.5)]
df[(df["dept"] == "Sales") | (df["dept"] == "IT")]
df[df["location"].isin(["Leeds", "London"])]
df[df["job_title"].str.contains("Manager")]
df[df["location"] != "London"]
df[~mask]                      # not
```

SORT

```
df.sort_values("hourly")                  # low to high
df.sort_values("hourly", ascending=False) # high to low
df.sort_values(["dept", "hourly"],
               ascending=[True, False])
df.nlargest(5, "hourly")   df.nsmallest(3, "hourly")
```

ROWS BY LABEL / POSITION

```
df.loc[5]            # row with index label 5
df.iloc[0]           # first row by position
df.iloc[-1]          # last row
df.loc[5, "annual_salary"]       # one cell
df.loc[mask, "bonus"] = 0        # set via mask
df = df.reset_index(drop=True)   # renumber 0..n
```

DATES

```
df["start"] = pd.to_datetime(df["start"])
df["start"].dt.year   .dt.month   .dt.day_name()
today = pd.Timestamp("2024-04-05")
(today - df["start"]).dt.days / 365.25
```

EXCEL → PANDAS

a sheet	DataFrame `df`
a column	Series `df["col"]`
AutoFilter	`df[mask]`
Sort A→Z	`.sort_values("col")`
helper column	`df["new"] = ...`
COUNTA / COUNTIF	`len(df)` / `mask.sum()`
Save As CSV	`.to_csv(..., index=False)`

GOTCHAS

- Column names are case-sensitive: `KeyError: 'Location'` means check the spelling
- Two columns need a list: `df[["a", "b"]]`, double brackets
- `sort_values` returns a copy; assign it or chain it
- Forgot `index=False`? Hello, "Unnamed: 0"
- Masks: `&` `|` `~` with brackets, never `and` `or` `not`